Lib-main

Reg. No. | | | | | | | | | | | [18ANE32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2021

ENGINEERING MATERIALS AND MANUFACTURING PROCESSES

Time: 03 Hours **Maximum Marks:70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Give five major classifications of materials based on the nature of materials.	**04**
		What are the major classifications of stainless steel?	**03**
	b.	List six important mechanical properties of engineering materials.	**04**
		List five important properties of titanium which make the metal popular in aerospace application.	**03**
2.	**a.**	What are the three major types of processing operations? What are the five major types of shaping operations?	**04**
		Give the expression for calculating the heat required to raise the temperature of metal to the pouring temperature. Define the terms used.	**03**
	b.	Distinguish between "pattern" and "core" used in metal casting.	**03**
		What are the four major types of patterns used for making moulds?	**02**
		Mention four major casting defects.	**02**
3.	**a.**	What are the three basic sheet metal-forming operations?	**03**
		Distinguish between "blanking" and "punching".	**02**
		What is "deep drawing"; "redrawing" and "reverse drawing"?	**02**
	b.	List the major defects encountered in deep drawing process.	**03**
		One cubic metre of a certain eutectic alloy is heated in a crucible from room temperature to 100 °C above its melting point for casting. The alloy's density is 7.5 g/cm^3, melting point is 800 °C, specific heat is 0.33 J/g.°C in the solid state and 0.29 J/g.°C in the liquid state; and heat of fusion is 160 J/g. How much heat energy must be added to accomplish the heating, assuming no losses?	**04**

OR

4.	**a.**	What are the two basic types of welding processes?	**01**
		List the major types of arc welding processes.	**03**
		How is heat generated in "thermit" welding. Illustrate the "thermit" welding process with a simple sketch.	**03**

	b.	What is "Friction Welding" and "Friction Stir Welding".	03
		What is flux cored-arc welding? Illustrate with a simple sketch.	04
5.	a.	What are the three major types of forging operations?	01
		Distinguish between forging hammer and forging press.	01
		Illustrate Impression-Die forging process with simple figures.	05
	b.	What are the two major components of forging machine?	02
		Write a short note on the different types of forging presses used in industry.	05

OR

6.	a.	Discuss five major terminology used in design of forging dies.	07
	b.	Illustrate roll forging with simple figures.	07
7.	a.	Differentiate between Bar Drawing and Wire Drawing.	01
		Write brief notes on:	06
		i. Thread rolling	
		ii. Gear rolling	
	b.	Distinguish between Cold-chamber and Hot-chamber die casting.	07

OR

8.	a.	What is direct extrusion? What is indirect extrusion?	04
		What is hot extrusion? What is cold extrusion?	03
	b.	The total solidification times of three casting shapes are to be compared: (1) a sphere with diameter is 10 cm, (2) a cylinder with diameter and length both 10 cm, and (3) a cube with each side is 10 cm. The same casting alloy is used in all the three cases.	07
		i. Determine the relative solidification times for each geometry.	
		ii. Based on the results of part (a), which geometric element would make the best riser?	
		iii. If the mould constant is 3.5 min/cm^2 in Chvorinov's rule, compute the total solidification time for each casting.	

Chvorinov's rule: $T_{TS} = C_m(V/A)^n$